When Random Splits Flatter Flood Forecasts: Three-Day High-Flow Warnings at Held-Out Indian Gauges

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***Abstract*—Machine-learning flood forecasters are often scored on random row splits, which let neighbouring days of the same river leak between training and test data. We build a leakage-controlled benchmark on CAMELS-IND for one question: will a gauge's flow exceed its 95th-percentile level within three days? We train on 193 gauges in 1980–2008, select on 2009–2014 and test once on 48 held-out gauges in 2015–2020; each test gauge still needs its own 1980–2008 record and a flow reading on the forecast day. Averaged over six gauge splits, a gradient-boosted model reached a balanced accuracy (BA) of 0.889; on the primary split it reached 0.898 (gauge-and-year bootstrap interval 0.870–0.916), against 0.750 for persistence. Scored on the same rows, a model trained on a random row split was 1.8 BA points higher (interval 1.2–2.5), so random splits overstate skill at new gauges. Recomputing the climate indices from training years only made no detectable difference, and restricting the model to inputs available in real time lowered BA by only 0.005. At the BA threshold, precision was only 0.328 and persistence had the higher critical success index (CSI); a rule that keeps persistence for rivers already above the level and uses the model elsewhere, set on validation, raised CSI to 0.500 against 0.448, at the cost of fewer onset warnings.**

***Index Terms*—CAMELS-IND, data leakage, early warning, flood forecasting, gradient boosting, streamflow**

I. Introduction

An emergency officer watching an Indian river gauge needs a plain answer: is the flow likely to reach its flood level in the next few days? Machine-learning models trained on many catchments now simulate streamflow well, even in basins without a gauge record [1]–[3]. Reported skill, however, depends on how a model is evaluated. When rows are split at random, days from the same flood fall on both sides of the split, and part of the test score then rewards memory of the event [4], [5].

India now has a large-sample dataset to study this properly. CAMELS-IND provides daily forcing, observed streamflow and catchment attributes for peninsular Indian catchments [6], in the format of the CAMELS family [7], [8]. It also ships a regional long short-term memory (LSTM) simulation of streamflow, which gives a strong deep-learning reference at no extra cost. What is missing is a short-range flood-exceedance benchmark on these gauges that controls leakage, compares against simple and strong references, and reports the numbers an operations room needs: alarm rates, onset lead time and calibration.

We build such a benchmark. The classifier is a standard gradient-boosted model; what is new is the task, the split and what they show:

1) A leakage-controlled exceedance benchmark on CAMELS-IND: gauges held out in space and time, thresholds and model choices from training and validation years only, repeated gauge splits and leave-basin-out folds, and a same-rows measurement of how much a random row split inflates the score (Fig. 1).
2) A feature-group ablation with paired gauge-bootstrap intervals, and comparisons with persistence, a trend rule, climatology and the CAMELS-IND LSTM. Today's flow carries most of the overall score; on rivers still below their level, removing the flow inputs made no detectable difference, while removing all the other inputs together did.
3) An operational evaluation: skill on below-level rows, alarms per 1,000 gauge-days, alarm episodes, calibration, a fallback that needs no flow reading, and a persistence-gated decision rule, chosen on validation, that beats persistence on the critical success index (CSI).

II. Related Work

Large-sample learning. The CAMELS datasets standardised forcing, flow and attributes across hundreds of basins [7], [8]. LSTMs trained on them learn regional behaviour [1], predict ungauged basins [2], simulate extremes well [9] and forecast extreme floods globally [3]; river-network graphs have not yet beaten them [10]. Most of this work simulates streamflow, and the exceedance on which warnings are issued is derived afterwards.

Operational forecasting. Physically based ensemble systems such as GloFAS [11] dominate operational forecasting [12], and a learned system now forecasts river stage for India and Bangladesh [13]. INDOFLOODS catalogues Indian flood events [14], and many studies fit tree ensembles or neural networks to single basins [15].

Evaluation. Leakage through the split is a leading cause of irreproducible machine-learning results [4], and data with spatial or temporal structure need blocked validation [5]. For rare events, precision–recall curves are more informative than ROC curves [16], and probabilistic forecasts should be checked for calibration [17].

Earlier studies held out basins for daily flow in the USA [2] or for flood events globally [3], held out time in Europe [10], or released CAMELS-IND with a regional LSTM but no exceedance task [6]. To our knowledge, no study has benchmarked short-range flood-exceedance forecasting on CAMELS-IND with a space-and-time held-out split against both a learned streamflow simulation and simple references.

III. Data and Task

A. Target

For gauge g on day t the target is

$$y_{g,t} = \mathbb{1}\Big[\max_{h\in\{1,2,3\}} Q_{g,t+h} \;\geq\; q_g^{95}\Big], \tag{1}$$

where Q is observed streamflow and q_g^{95} is the 95th percentile of the gauge's flow over the training years 1980–2008 only. A day with $y_{g,t} = 1$ is a *flood window*. The level is a statistical proxy, exceeded on one day in twenty in the training years; CAMELS-IND gives no official danger levels, and a stricter 99th-percentile level is tested in Section V-D. We assume that three days is enough to stage rescue teams; we did not survey operators. Days with no flow observation on any of $t{+}1$ to $t{+}3$ are dropped, and when only some are observed the maximum is taken over those, which can miss an exceedance on a missing day. The 281,360 dropped gauge-days (9.3%) were not concentrated in the monsoon (26% against 34% of kept days) or after high flow.

B. Inputs

The 50 inputs, all known at the end of day t, are 11 streamflow features (lags of 0–3 days, 1- and 3-day changes, 7- and 30-day means, 7-day maximum, log flow and a reading flag), 10 rainfall features (lags, 3- to 90-day sums, 7-day maximum), 5 weather and 6 soil-moisture inputs, 2 for the season and 16 catchment attributes (area, slope, climate indices, land cover, soils). All flow features except one log-flow input are scaled by q_g^{95}, so they read as "how close the river is to its flood level". Hydrological signatures of CAMELS-IND are excluded, because they are computed over the full 1980–2020 record and would leak test-period information; future rainfall is excluded as well. The four CAMELS-IND climate indices we keep (mean rainfall and evapotranspiration, aridity, high-rainfall frequency) are computed over the dataset period, which includes our test years; recomputed from 1980–2008 forcing only, they gave the same skill (Section V-D). Rainfall is the gridded product of the India Meteorological Department (IMD); soil moisture, humidity and evapotranspiration come from the Indian Monsoon Data Assimilation and Analysis (IMDAA) reanalysis and the Global Land Evaporation Amsterdam Model (GLEAM) [6]. These products are released with a delay, so Section V-D also tests the model without them.

C. Split

Fig. 1 shows the split. All 242 gauges in the observed-streamflow file have forcing data and at least three years of flow in 1980–2008, so none is excluded; the dataset paper describes 228 catchments [6], while the streamflow file we use lists 242, all with more than 30% flow availability. This is not an ungauged-basin method: each test gauge needs its 1980–2008 record for the flood level, climatology and flow scaling, and a flow reading on day t; "unseen" means unseen in training. A group split puts 20% of gauges aside (seed 42), so the primary test asks about rivers the model has never seen, in years it has never seen; of the 49 unseen gauges, 1 gauge has no flow record in 2015–2020, which leaves 48 test gauges. Training uses the other gauges in 1980–2008 (1,641,197 gauge-days), validation the same gauges in 2009–2014 (349,645), and the test the unseen gauges in 2015–2020, all labelled by (1) (52,994 gauge-days, of which 5.5% are flood windows).

seen gauges	Train 1980–2008	Validation 2009–14	not used
unseen gauges	not used (threshold, climatology)	not used	**Test** 2015–20

Fig. 1. Leakage-controlled split. The primary test uses gauges and years that played no part in training, model selection or threshold choice. The unseen gauges' 1980–2008 record is used only to set their flood level and climatology.

IV. Methods

Model. Seven candidates (logistic regression, random forest, histogram gradient boosting and four XGBoost [18] configurations) were compared on validation balanced accuracy, and the decision threshold of the selected depth-6 XGBoost model, 0.45, was set on validation as well. The selected model uses a learning rate of 0.05, subsampling of rows and columns at 0.8, a minimum child weight of 20, L2 regularisation of 5 and a positive-class weight equal to the ratio of negative to positive training rows. Training stops early on the validation area under the precision–recall curve (PR-AUC). The candidates were fitted on one 500,000-row training sample; the depth-6 configuration was then refit on all training rows and kept because it scored higher on validation. In validation balanced accuracy all seven lay between 0.899 (logistic regression) and 0.911 (the selected model). The sample-fitted model had been scored on the test set during development, before this refit; no choice of model, input or threshold used that score, and the later six-split and basin-fold results agree. Balanced accuracy is $\mathrm{BA} = (\mathrm{TPR} + \mathrm{TNR})/2$, the mean of the true positive and true negative rates.

References. Persistence forecasts a flood window if the river is above its flood level today, and no flood on a day without a reading. A *trend rule* needs no learning: its score is $Q_{g,t}/q_g^{95} + \max(0, Q_{g,t} - Q_{g,t-3})/q_g^{95}$, so a high or rising river raises an alarm. *Climatology* forecasts the gauge's own 1980–2008 rate of flood windows within ± 15 days of the calendar day. The *LSTM simulation* is the regional LSTM streamflow of CAMELS-IND [6], scored as $\max_{h\leq 3} \hat{Q}_{g,t+h}/q_g^{95}$. It is driven by observed forcing, so it sees the rain of days $t{+}1$ to $t{+}3$, which no real forecast has. It was trained by the dataset authors on 159 catchments over 1991–2015 and tested on 2016–2020 [6], so it has most likely seen our test gauges and, in 2015,

one of our test years. It is therefore an optimistic reference, not a held-out competitor. Thresholds of the references were set on validation like the model's.

Analyses. We refit the selected configuration on the same 500,000-row training sample with one input group removed at a time, and with only the streamflow and season inputs or only the meteorological and season inputs. Percentile intervals come from 1,000 resamples of the test gauges (500 for the protocol comparison and the ECE), which keeps each river's days together; the main score also gets a gauge-by-year resample, because the test covers few years. Calibration is the expected calibration error (ECE) [19] over ten bins, before and after isotonic regression fitted on the validation years of the seen gauges and applied unchanged to the unseen test gauges. A flood onset is a day on which the river reaches q_g^{95} after at least seven days below it; a method warns of it if it forecasts a flood window one, two or three days earlier. The monsoon is June to September. Robustness refits (Section V-D) use the ablation's 500,000-row sample. They cover five more random gauge splits (seeds 1 to 5), a leave-basin-out evaluation and the same task with the 99th percentile as the level. For leave-basin-out, the CAMELS-IND river basins are grouped into five folds, and every basin is held out once and tested on 2015–2020.

Reproducibility. Splits and resamples use seed 42; code, model and the list of held-out gauges are available from the authors (link withheld for review), and every number here is generated from the experiment outputs by a script that checks each claim.

V. Results

A. Skill on Unseen Gauges

Table I and Fig. 2 give the main comparison. The model scored above persistence and climatology in balanced accuracy, recall and PR-AUC, and its interval for the gain over persistence (0.123–0.175) excludes zero. Resampling test years as well as gauges widened the interval of its balanced accuracy to 0.870–0.916. Persistence was more precise (0.789 against 0.328) because a river already above its level usually stays there, but it misses every flood that has not yet begun. On below-level rows, where the river is still below the level, the model reached a balanced accuracy of 0.857 and a PR-AUC of 0.369, above the trend rule, climatology and persistence. On the critical success index, $\mathrm{CSI} = \mathrm{TP}/(\mathrm{TP} + \mathrm{FP} + \mathrm{FN})$ [20], which ignores correct negatives, persistence scored highest at this threshold (0.448, against 0.317 for the model); Section V-F returns to this.

The LSTM simulation is a demanding reference because it knows the rain that is still to fall. Over all rows its balanced accuracy was 0.873 against 0.898 for the model, and the paired gauge-bootstrap interval of the difference, -0.009 to 0.063, includes zero. On below-level rows the LSTM scored higher (0.877 against 0.857), as its access to future rain would suggest, yet this gap was also within sampling noise (interval -0.045 to 0.011).

TABLE I
Three-day exceedance on 48 unseen gauges, 2015–2020

Method	BA	95% CI	all rows PR-AUC	Rec.	Prec.	CSI	below-level BA
XGBoost (ours)	0.898	0.886–0.910	0.737	0.902	0.328	0.317	0.857
No-flow XGB.‡	0.853	0.820–0.880	0.494	0.865	0.239	0.230	0.846
LSTM sim.†	0.873	0.829–0.911	0.589	0.826	0.376	0.349	0.877
Trend rule	0.828	0.811–0.845	0.605	0.754	0.305	0.277	0.734
Climatology	0.802	0.774–0.829	0.203	0.891	0.152	0.149	0.802
Persistence	0.750	0.724–0.778	0.428	0.508	0.789	0.448	0.500

CI: 95% gauge bootstrap. Below-level: river below q_g^{95} on day t.
†Sees observed future rain. ‡500,000-row refit (Table III).

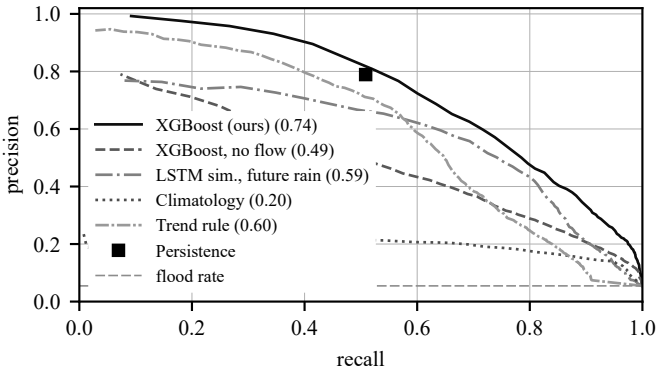


Fig. 2. Precision–recall curves on the unseen test gauges; PR-AUC in brackets. The square is persistence, a single operating point; the flat line is the flood rate, the level of no skill.

B. Effect of the Evaluation Protocol

To see what the split alone does, we trained the same configuration under four protocols, each with its own validation set, and scored each once (Table II). The random row split scored highest, but its test set differs from ours in gauges, years and flood rate. We therefore also scored it and the held-out model on the same 10,793 rows: random-split test rows that belong to held-out gauges in 2015–2020. On these shared rows the random split was 1.8 points higher in balanced accuracy (interval 1.2–2.5) and 3.2 points higher in PR-AUC (1.8–4.7). It had trained on other days of the same rivers and years, the most likely reason. Holding out later years lowered PR-AUC more than holding out gauges, which suggests that a temporal split matters at least as much as a spatial one, although flood rates differ between those test sets.

C. What the Forecast Relies On

Over all rows, removing the streamflow inputs cost the most in the ablation (Table III), lowering balanced accuracy from 0.898 to 0.853 and PR-AUC from 0.736 to 0.494, while removing soil moisture, rainfall or catchment attributes cost little. On below-level rows the picture changes. Without streamflow the below-level balanced accuracy fell only from 0.852 to 0.846 (paired difference -0.005 to 0.020), whereas streamflow and season alone reached 0.810 (difference 0.025 to 0.061) and meteorology and season alone 0.825. For rivers still below their level, then, the non-flow inputs carried the

TABLE II
SAME CONFIGURATION, FOUR PROTOCOLS (500,000-ROW REFIT)

Test set	Gauges	Flood %	BA	95% CI	PR-AUC	Prec.
Random rows	242	7.4	0.915	0.911–0.919	0.810	0.426
New years	181	6.1	0.892	0.885–0.898	0.724	0.349
New gauges	49	7.6	0.905	0.898–0.912	0.789	0.424
New gauges + years	48	5.5	0.898	0.886–0.911	0.736	0.340

Random: 70/10/20% of gauge-days. Others: trained on seen gauges, 1980–2008; new years: seen gauges with flow in 2015–20; new gauges: unseen, 1980–2008 (threshold years). CIs rounded; at four decimals the first and last do not overlap.

TABLE III
FEATURE-GROUP ABLATION (500,000-ROW TRAINING SAMPLE)

Inputs	n	BA	PR-AUC	Rec.	Below-level BA
All inputs	50	0.898	0.736	0.897	0.852
No streamflow	39	0.853	0.494	0.865	0.846
No catchment	34	0.895	0.734	0.907	0.855
No soil moisture	44	0.896	0.732	0.895	0.854
No rainfall	40	0.897	0.730	0.901	0.857
Streamflow + season	13	0.875	0.685	0.882	0.810
Meteorology + season	23	0.832	0.447	0.833	0.825

Below-level BA: rows where the river is below its flood level on day t.

skill and the flow inputs added little, although neither group alone matched the full model. The non-flow groups overlap: removing rainfall, soil moisture or catchment attributes one at a time changed below-level balanced accuracy by less than 0.006, so these single-group results cannot rank them. The no-flow variant also scored above climatology overall (0.853 against 0.802), so it is a usable fallback when a gauge reading is missing.

D. Robustness

Table IV repeats the evaluation with refits that are directly comparable with each other. The primary split sits at the top of the range over six gauge splits, so its scores are, if anything, optimistic; averaged over all six, balanced accuracy was 0.889 (s.d. 0.008). Whole river basins held out gave 0.866–0.906, so gauges on the same river as a training gauge did not carry the result. With the stricter 99th-percentile level, only 1.6% of test rows were positive and balanced accuracy fell to 0.853. By year, skill was steady over 2015–2018 but lower in the sparse late years (0.783 in 2019 and 0.767 in 2020).

The inputs passed three checks. Climate indices recomputed from training years only, or dropped altogether, made no detectable difference. Dropping the late-arriving reanalysis inputs (humidity, evapotranspiration and soil moisture) lowered balanced accuracy by only 0.005 (interval 0.000–0.011) and below-level skill by no detectable amount. Two other learners did worse on the same sample: logistic regression came close in balanced accuracy but trailed by 5.4 points of PR-AUC, and a neural network (a multilayer perceptron with a 14-day input history) reached 0.862, below the model by 0.023–0.050.

TABLE IV
ROBUSTNESS REFITS (500,000-ROW TRAINING SAMPLE)

Evaluation	Gauges	BA	PR-AUC	Below-level BA
Primary split	48	0.898	0.736	0.852
5 other gauge splits	45–47	0.877–0.898	0.706–0.764	0.822–0.851
5 basin folds	45–47	0.866–0.906	0.648–0.829	0.785–0.864
99th-percentile level*	48	0.853	0.364	0.798
Climate indices, 1980–2008	48	0.897	0.737	0.855
No climate indices	48	0.896	0.729	0.854
Real-time inputs only	48	0.893	0.721	0.844
Logistic regression	48	0.893	0.682	0.843
MLP, 14-day history	48	0.862	0.596	0.822

Ranges over splits or folds. Below: river below its level on day t.
*1.6% positives, so its PR-AUC is not comparable.

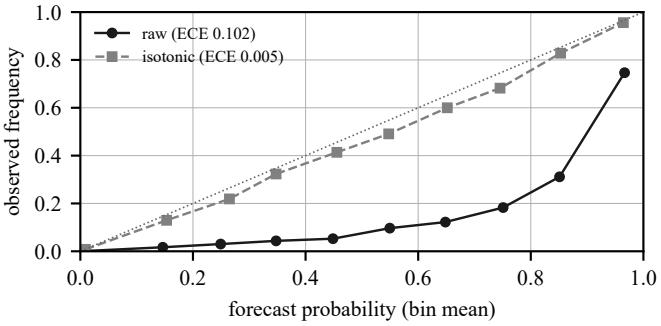


Fig. 3. Reliability of the forecast probabilities on the unseen test gauges, before and after isotonic recalibration fitted on validation data (bins with at least 50 gauge-days).

E. Operational Behaviour

Precision is the main weakness. At the validation threshold the model raised 150 alarms per 1,000 gauge-days, and only 0.328 of them were followed by a flood window; raising the threshold to 0.75 halved the alarms and lifted precision to 0.539.

The raw probabilities were also over-confident (ECE 0.102), mostly in the monsoon (0.250 against 0.026 in other months), and their Brier score was worse than climatology's (0.065 against 0.048). Isotonic recalibration [21] fitted on validation data brought the ECE down to 0.005 (Fig. 3), so the recalibrated score, not the raw one, can be shown to an officer as a probability. When the model was refit without the positive-class weight, its raw ECE was 0.008, so the weighting explains most of the over-confidence. That refit, with its own threshold chosen on validation, was at least as good in balanced accuracy (0.902 against 0.898), which suggests, from test data only, that the unweighted model would serve deployment better; we report the weighted one because it was selected on validation before the test was scored.

The model first warned three days ahead of 63.9% of the 266 onsets at 44 gauges. Climatology and the LSTM simulation did better on this measure, the first because it alarms on most monsoon days, the second because it sees future rain. Lead counts alone reward constant alarms: independent

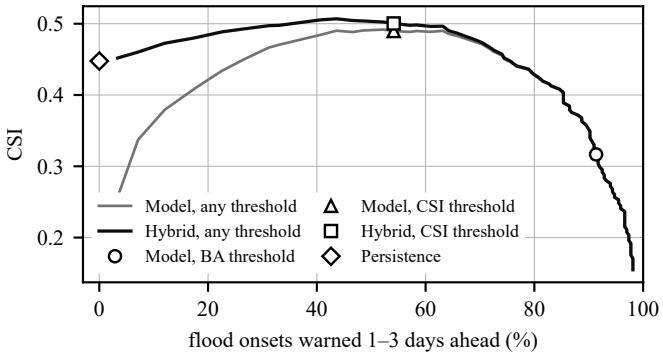


Fig. 4. Critical success index against the share of flood onsets warned one to three days ahead, on the unseen test gauges, for every threshold. Markers: operating points with thresholds chosen on validation. Hybrid: persistence when the river is above its level, otherwise the model.

random alarms at the model's rate would warn of 38.6% of onsets on at least one of the three days before them, against 91.4% for the model. Officers act on episodes rather than days, and of the model's 1,441 alarm episodes 77% contained no flood window.

Skill was consistent across rivers: over the 46 test gauges with at least one flood window, the median per-gauge balanced accuracy was 0.900 (interquartile range 0.860–0.926; worst 0.769). Counting any warning one to three days ahead, the model caught 91.4% of the onsets. The no-flow variant caught more (94.0%), but only because it raised more alarms (197 per 1,000 gauge-days): in the 500,000-row refits at a common alarm rate, it warned of 88.7% against 89.5% for all inputs.

F. Operating Rules

The threshold above maximises balanced accuracy, which favours recall. For committing teams, CSI is the more usual score [20], so we also chose thresholds that maximise CSI on the validation years and scored each rule once on the test set (Fig. 4). At its CSI threshold the model alone beat persistence (CSI 0.490 against 0.448; gain 0.019–0.065). A hybrid rule that raises an alarm whenever the river is already above its level, and otherwise uses the model, added a small further gain (CSI 0.500; gain over the model 0.004–0.020). It cut the alarms from 150 to 56 per 1,000 gauge-days and raised precision to 0.658, but warned of 54% of the flood onsets instead of 91%. The trade holds along the whole curve: every threshold that warned of at least 80% of onsets had a CSI of at most 0.424. Separate monsoon and dry-season thresholds did not help. The BA threshold therefore suits an early review list, and the hybrid rule suits committing teams.

G. Where the Forecast Succeeds and Fails

Lead day. The target pools three lead days. Recall fell from 96.7% for a flood on day $t{+}1$ to 89.5% on day $t{+}3$, and from 90.3% to 80.0% on rivers still below their level on day t.

Season. In June to September the model raised 372 alarms per 1,000 gauge-days at a precision of 0.311, against 36 and 0.415 in the rest of the year, and 86% of all false alarms fell in the monsoon. Most of these were not near misses: in the median monsoon false alarm the river peaked at 41% of its flood level over the next three days, and only 21% of them reached 70–100%.

Catchment. Over the 46 test gauges with flood windows, per-gauge balanced accuracy rose with catchment area (Spearman $\rho = 0.44$, $p = 0.002$), which is consistent with large rivers rising slowly enough for today's flow to describe the next three days. Aridity, mean rainfall and slope showed no relation ($p \geq 0.64$).

VI. Discussion and Limitations

Three findings matter for practice. First, the inputs play different roles. Today's flow carries the overall score, but skill on rivers still below their level owed little to it, so for new floods a meteorological feed matters as much as a gauge feed; when the gauge feed fails, the no-flow variant is a better fallback than climatology. Second, precision is the weak point, mainly in the monsoon, and even the LSTM with future rain reached only 0.376. The hybrid rule raises precision but warns of fewer onsets, so a positive forecast should start an officer's review, and an evacuation needs further evidence. Third, scored on the same rows, a random row split raised balanced accuracy by 1.8 points and PR-AUC by 3.2 points, so published Indian flood models can only be compared under the same split.

Studies of Indian rivers should therefore hold out gauges and later years, take every threshold from training and validation years, and report precision, alarm rates, below-level skill and calibration.

The study has limits. Almost all test gauge-days fall in 2015–2018, because CAMELS-IND holds few flow records for 2019 and 2020. The gauges are in peninsular India only, so skill may not transfer to Himalayan or Brahmaputra rivers. The 95th-percentile level is a statistical proxy, not an official danger level. Inputs available in real time cost little skill, but we used observed rainfall and no rainfall forecast. The LSTM reference was not held out: it was trained on years and, most likely, gauges that overlap our test set, and it sees future rain, so the comparison is indicative only; our own neural baseline was a multilayer perceptron, not a sequence model such as an LSTM trained on the same split. We pool three lead days into one target, and the ablation and protocol models were refitted on a 500,000-row sample (all-input refit 0.898, against 0.898 for the model trained on all rows).

VII. Conclusion

On CAMELS-IND gauges held out in space and time, a gradient-boosted model forecast three-day exceedance of the 95th-percentile flow with a balanced accuracy of 0.889 on average over six gauge splits and 0.898 on the primary split. It scored above persistence, the trend rule and climatology, and it held up over held-out basins (0.866–0.906). Scored on the same rows, a random row split overstated balanced accuracy by 1.8 points. Climate indices from training years gave the same skill, and inputs available in real time lowered balanced

accuracy by only 0.005. On rivers still below their level the skill was lower (0.857) and owed little to the flow inputs. At the balanced-accuracy threshold most alarms were false, but a rule that keeps persistence for rivers already above the level and uses the model elsewhere beat persistence on CSI (0.500 against 0.448), though only slightly better than a CSI threshold on the model alone and at the cost of fewer onset warnings. A positive forecast should start an officer's review; the split, the protocol and the two decision rules give a reference against which other forecasters on Indian rivers can be compared.

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